

Bitwise differential trail through **LBlock** found by SAT

CiVerLy

July 24, 2026

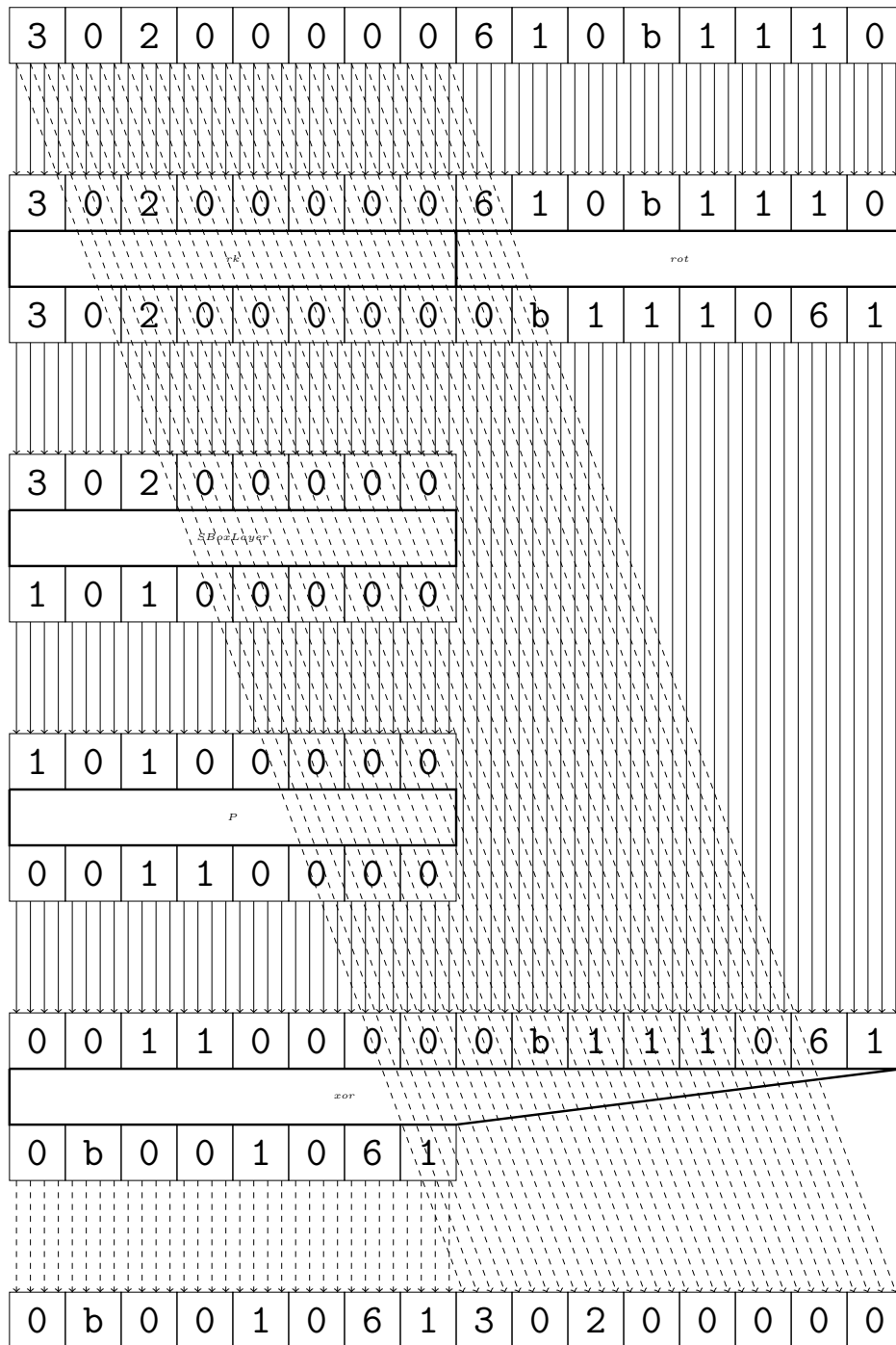
1 LBlock

Maximal differential probability: 2^{-48}

3	0	2	0	0	0	0	6	1	0	b	1	1	1	0
3	0	2	0	0	0	0	6	1	0	b	1	1	1	0
ShiftRow														
0	b	0	0	1	0	6	1	3	0	2	0	0	0	0
0	b	0	0	1	0	6	1	3	0	2	0	0	0	0
ShiftRow														
0	0	0	0	0	1	0	1	0	b	0	0	1	0	6
0	0	0	0	0	1	0	1	0	b	0	0	1	0	6
ShiftRow														
0	0	1	0	0	0	0	b	0	0	0	0	0	1	0
0	0	1	0	0	0	0	b	0	0	0	0	0	1	0
ShiftRow														
0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
ShiftRow														
0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
1	0	0	0	0	b	0	0	0	0	0	0	0	0	0
1	0	0	0	0	b	0	0	0	0	0	0	0	0	0
ShiftRow														
0	0	a	0	2	0	0	0	1	0	0	0	0	b	0
0	0	a	0	2	0	0	0	1	0	0	0	0	b	0
ShiftRow														
0	0	0	0	0	0	0	0	0	0	a	0	2	0	0
0	0	0	0	0	0	0	0	0	0	a	0	2	0	0
ShiftRow														
a	0	2	0	0	0	0	0	0	0	0	0	0	0	0
a	0	2	0	0	0	0	0	0	0	0	0	0	0	0
ShiftRow														
0	0	2	1	0	0	0	a	0	2	0	0	0	0	0
0	0	2	1	0	0	0	a	0	2	0	0	0	0	0
ShiftRow														
2	1	0	1	0	0	a	0	0	0	2	1	0	0	0
2	1	0	1	0	0	a	0	0	0	2	1	0	0	0
ShiftRow														
0	0	2	0	0	0	0	2	2	1	0	1	0	0	a
0	0	2	0	0	0	0	2	2	1	0	1	0	0	a
ShiftRow														
0	1	0	1	a	1	2	1	0	0	2	0	0	0	2
0	1	0	1	a	1	2	1	0	0	2	0	0	0	2
FinalKey														
0	0	2	0	0	0	0	2	0	1	0	1	a	1	2
0	0	2	0	0	0	0	2	0	1	0	1	a	1	2

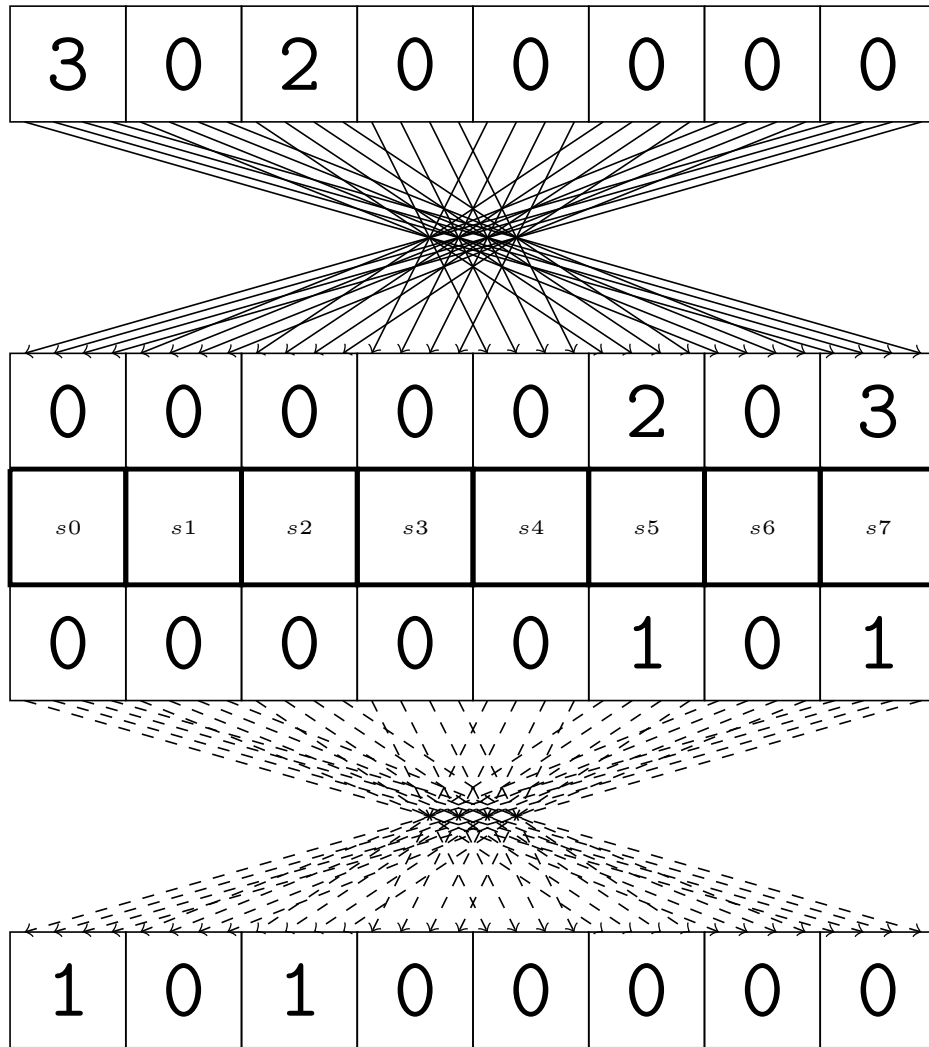
2 LBlock_round

Maximal differential probability: $2^{-4.0}$

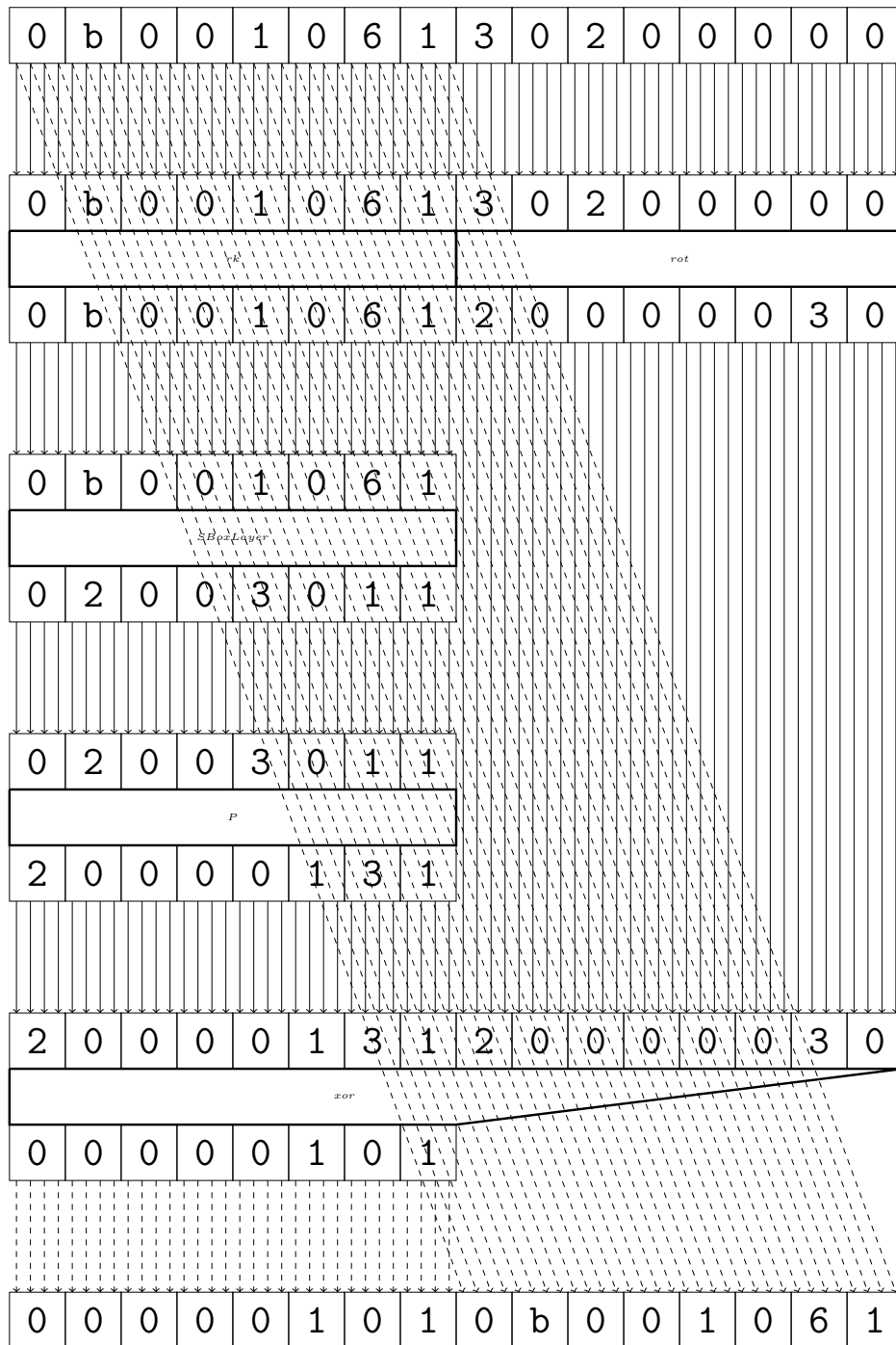


3 SBoxLayer

Maximal differential probability: $2^{-4.0}$

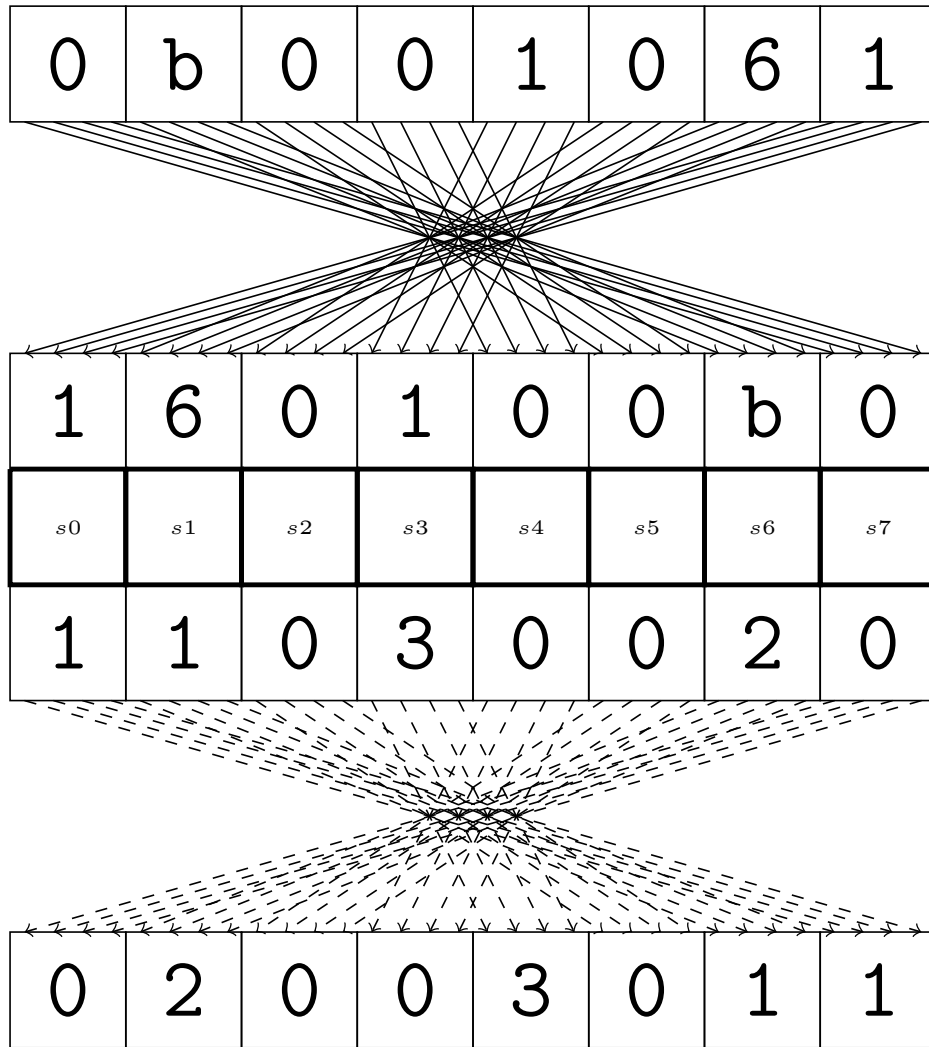


4 LBlock_round

Maximal differential probability: $2^{-8.0}$ 

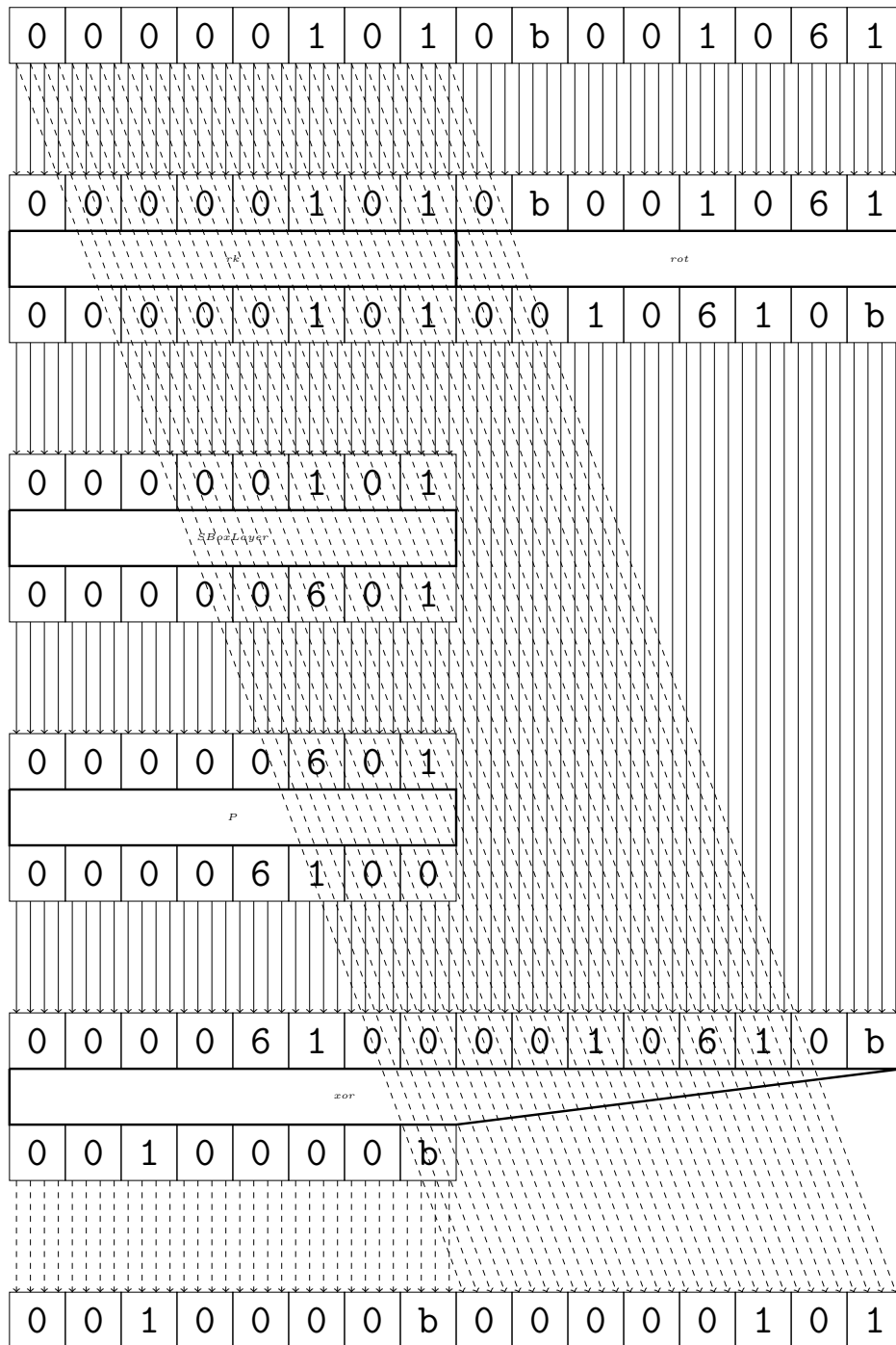
5 SBoxLayer

Maximal differential probability: $2^{-8.0}$



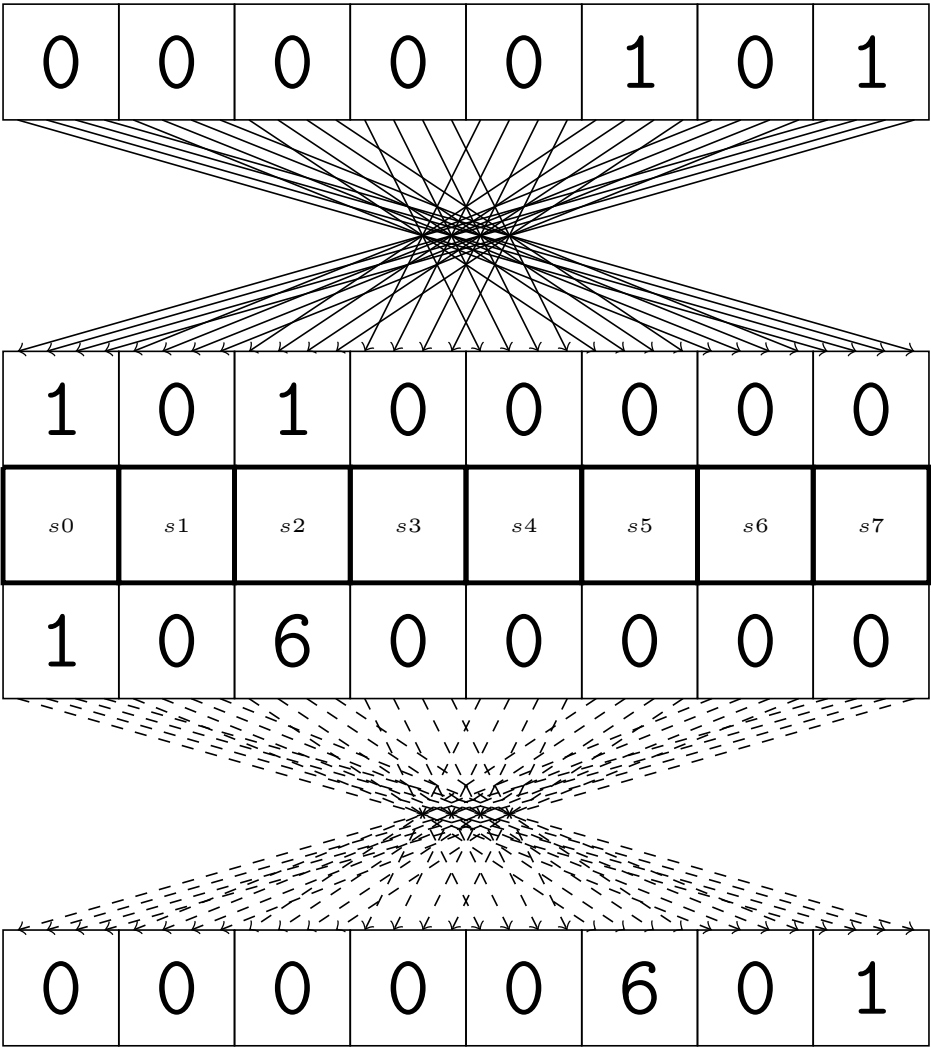
6 LBlock_round

Maximal differential probability: $2^{-4.0}$



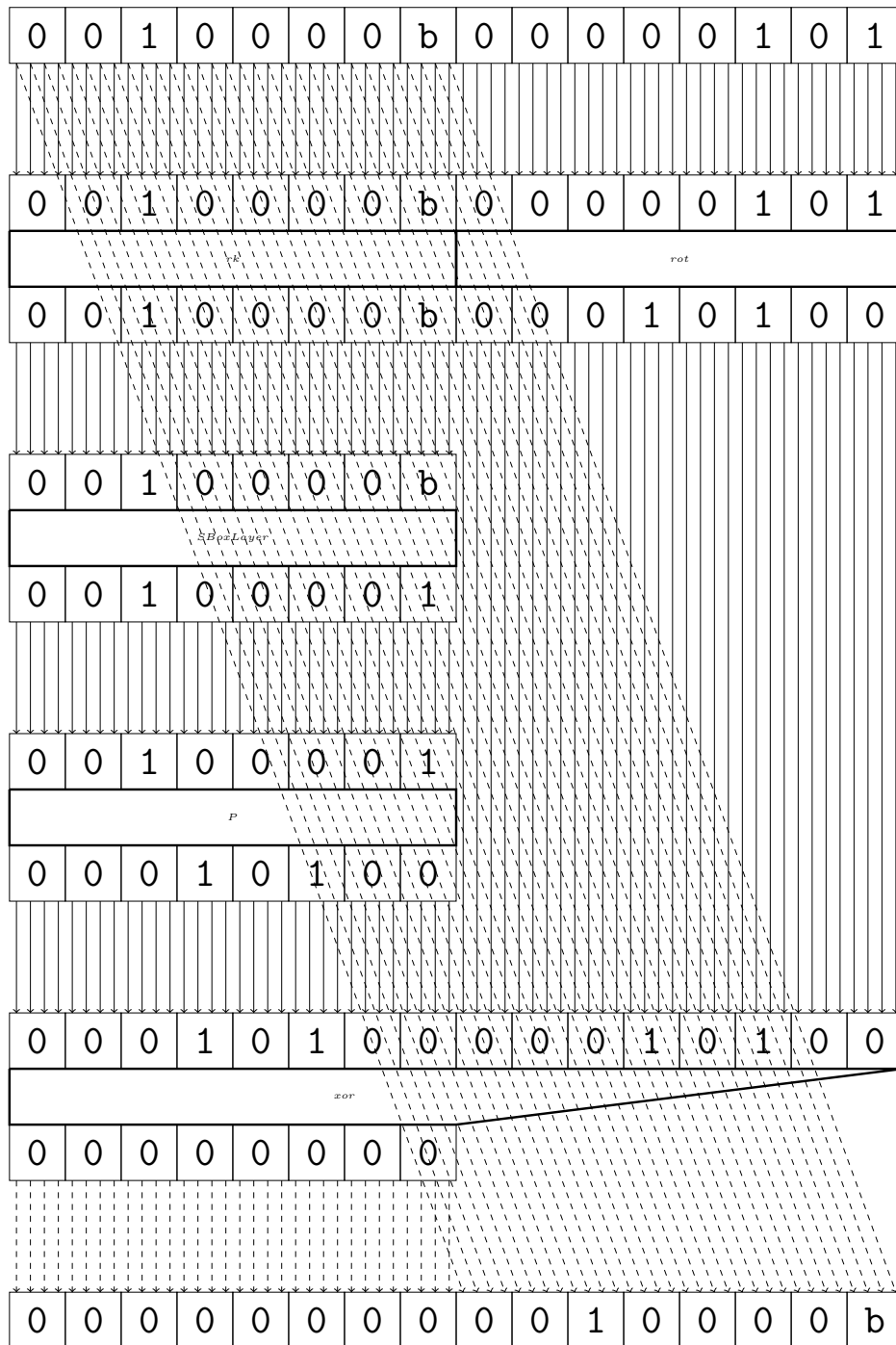
7 SBoxLayer

Maximal differential probability: $2^{-4.0}$



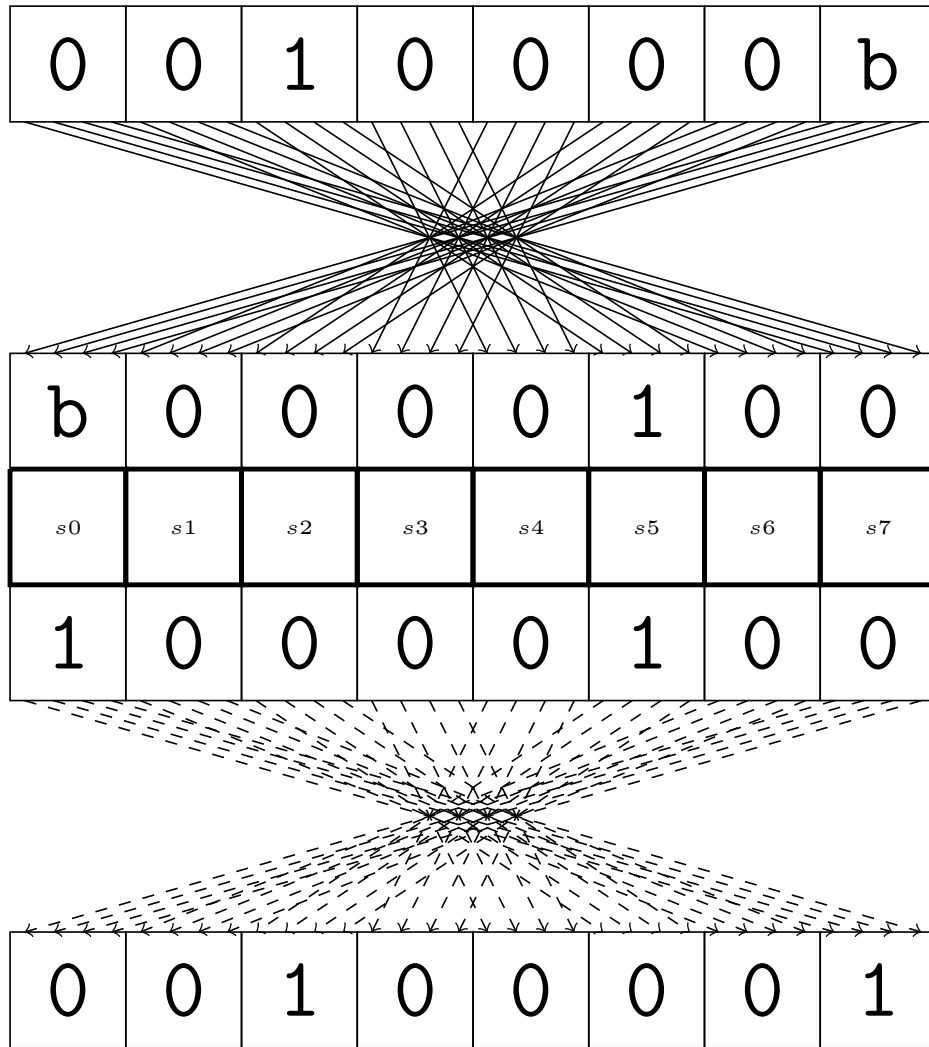
8 LBlock_round

Maximal differential probability: $2^{-4.0}$



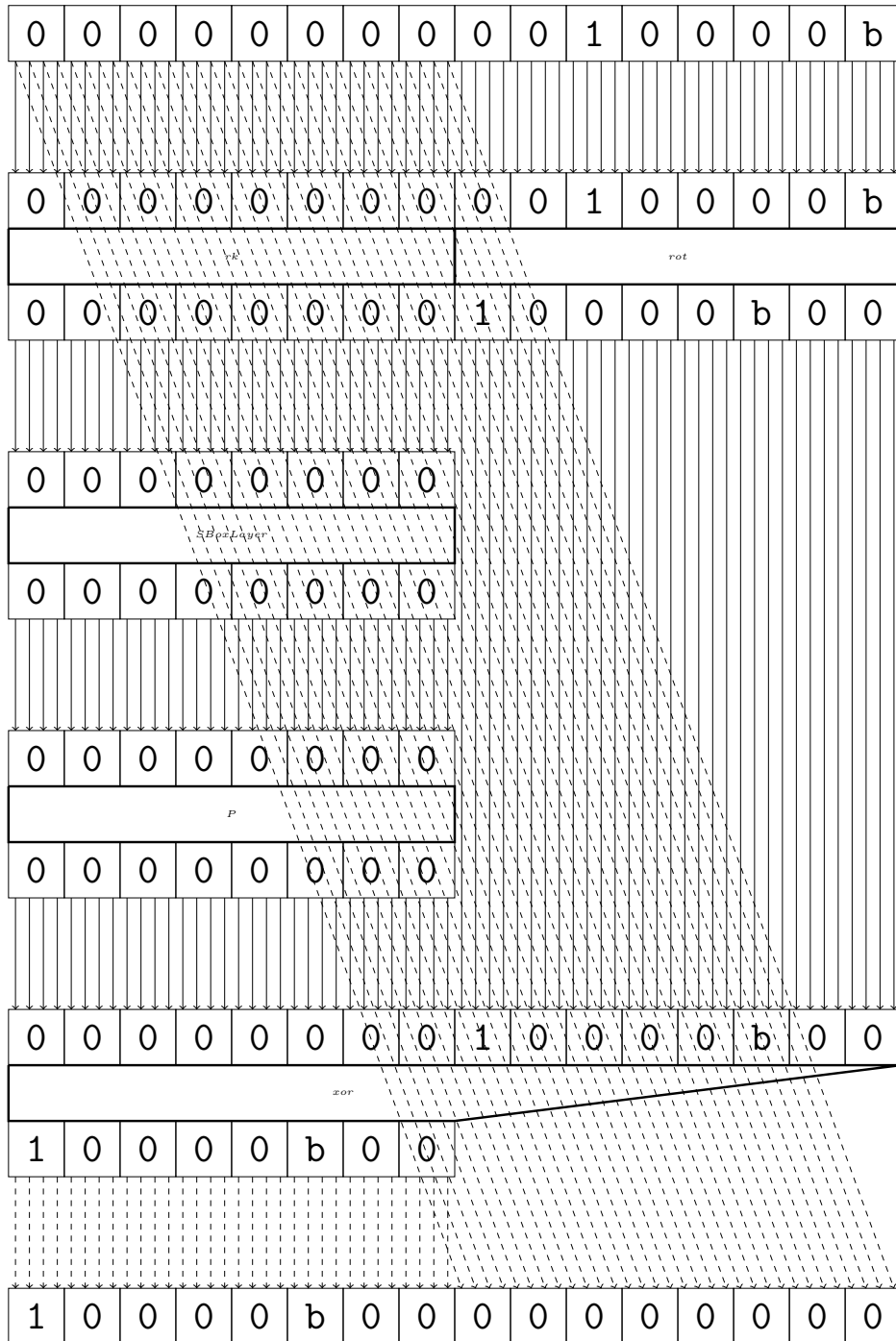
9 SBoxLayer

Maximal differential probability: $2^{-4.0}$



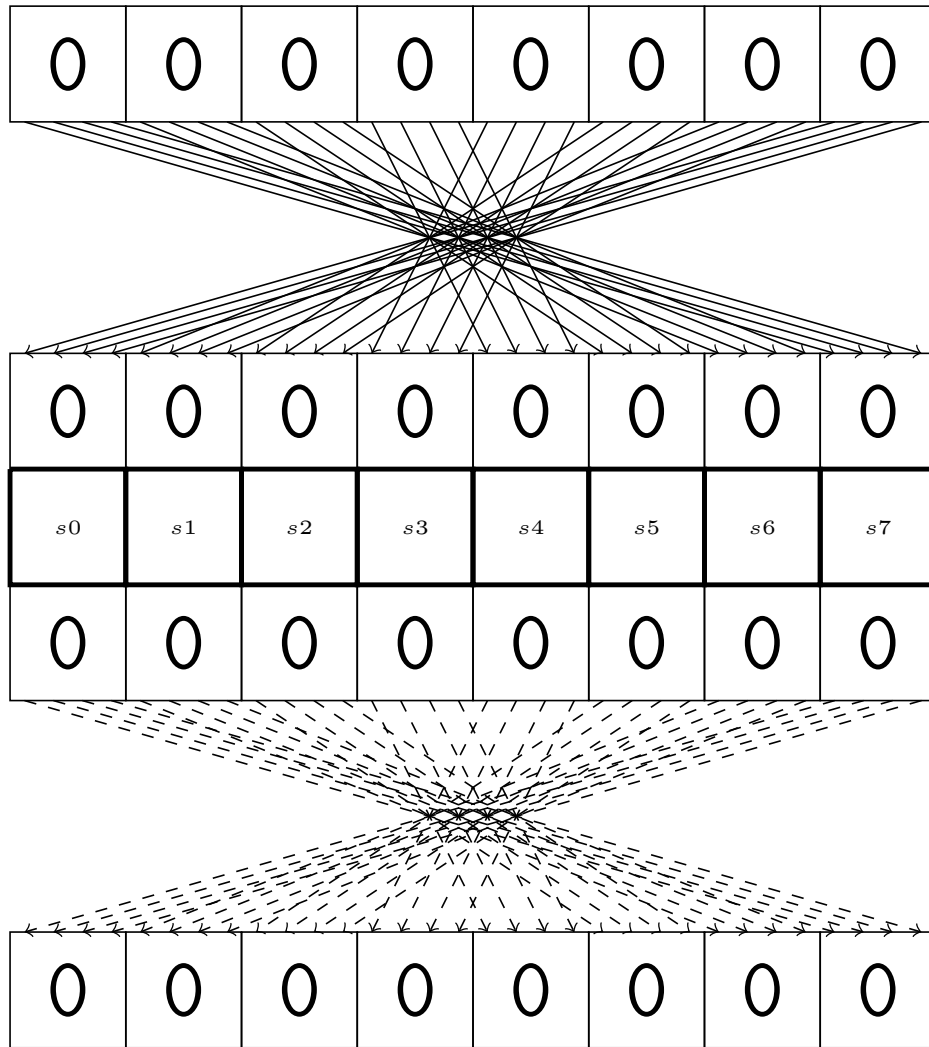
10 LBlock_round

Maximal differential probability: $2^{-0.0}$



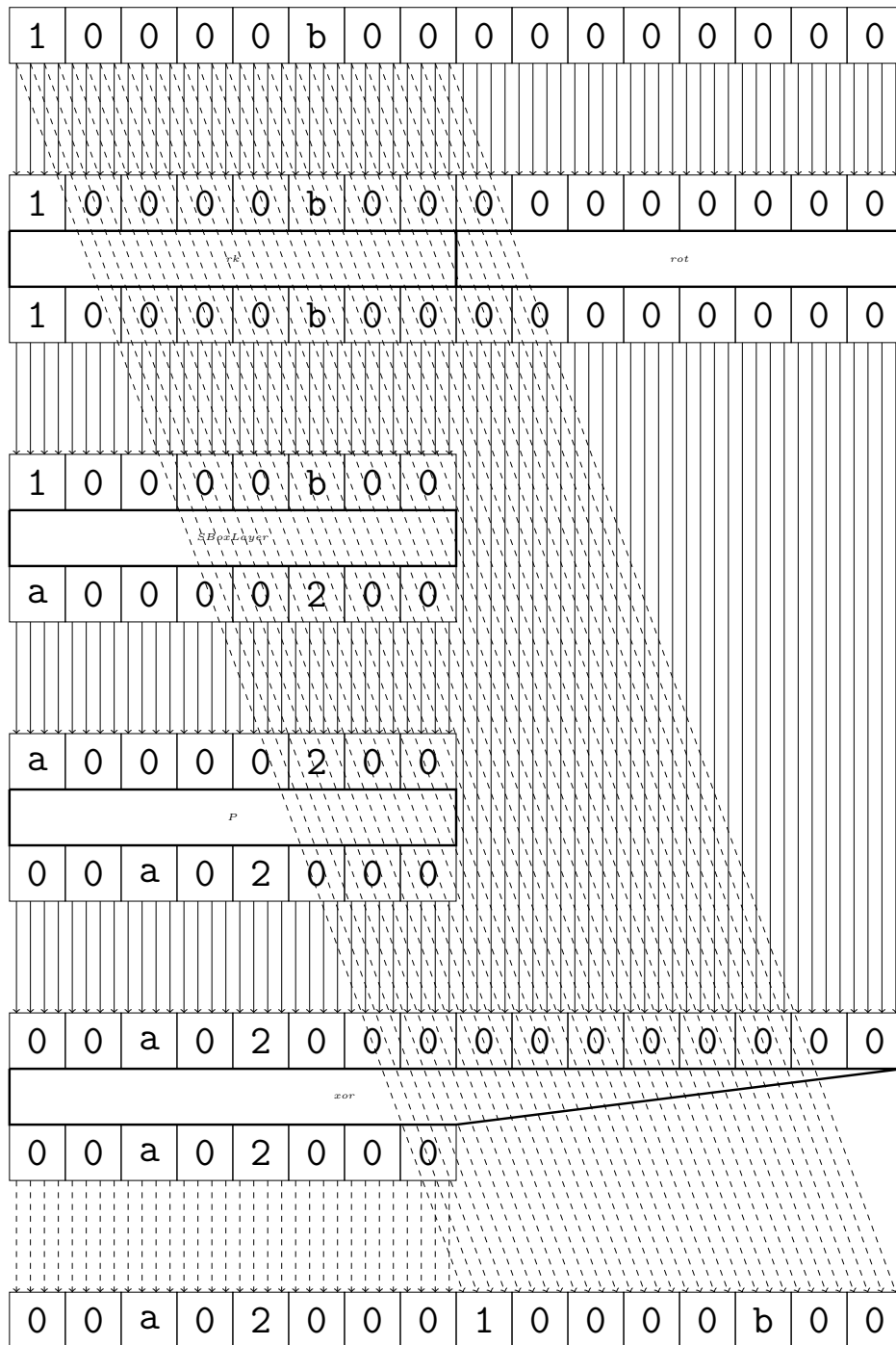
11 SBoxLayer

Maximal differential probability: $2^{-0.0}$



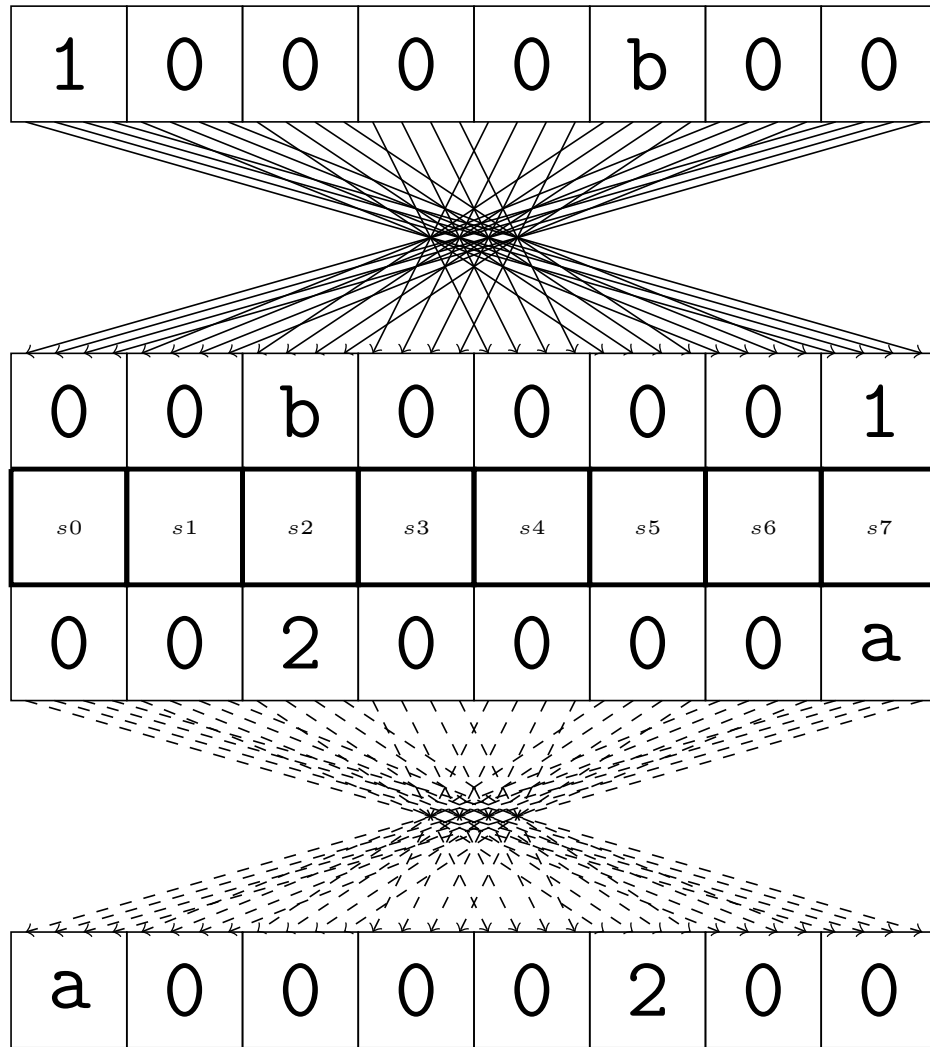
12 LBlock_round

Maximal differential probability: $2^{-4.0}$

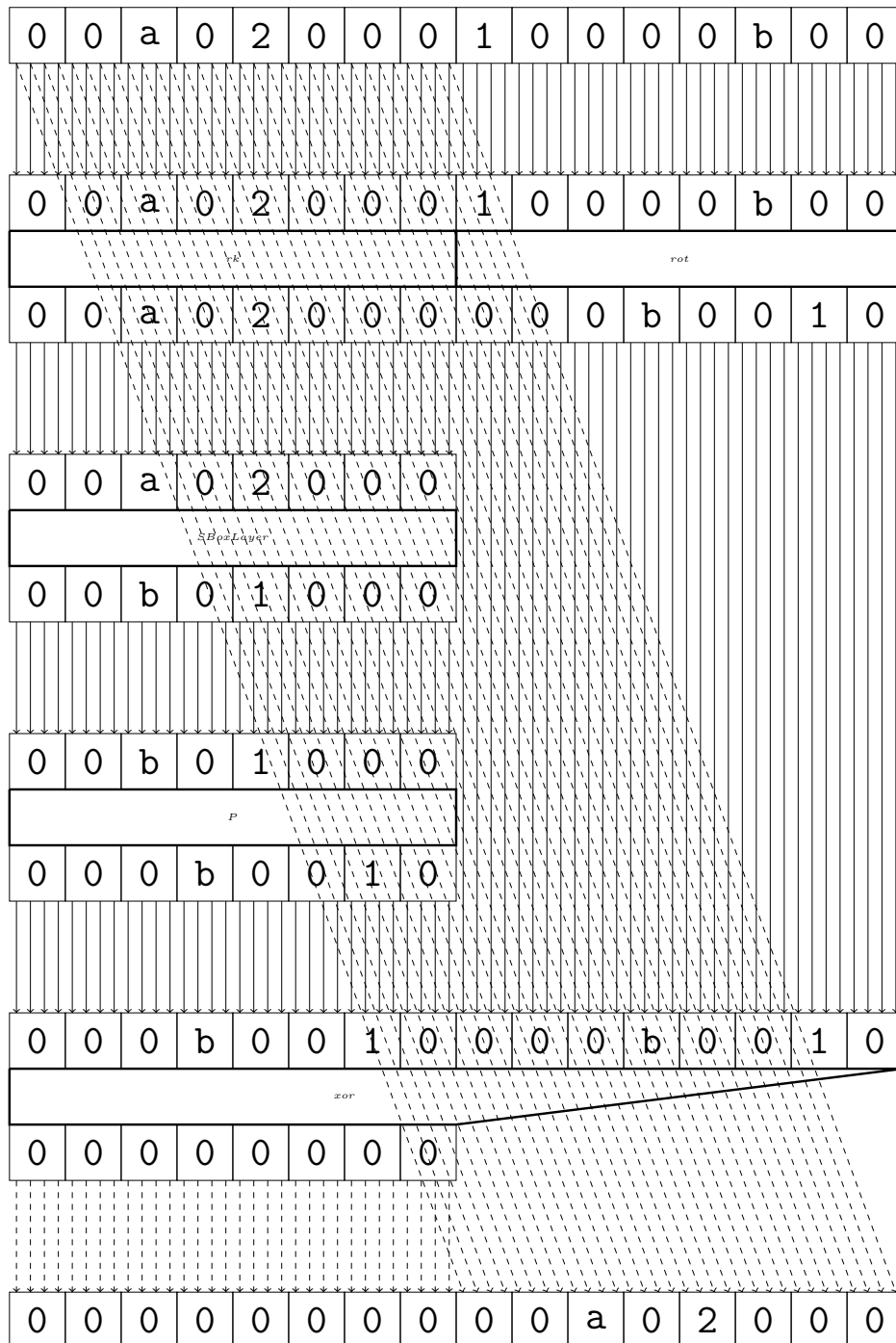


13 SBoxLayer

Maximal differential probability: $2^{-4.0}$

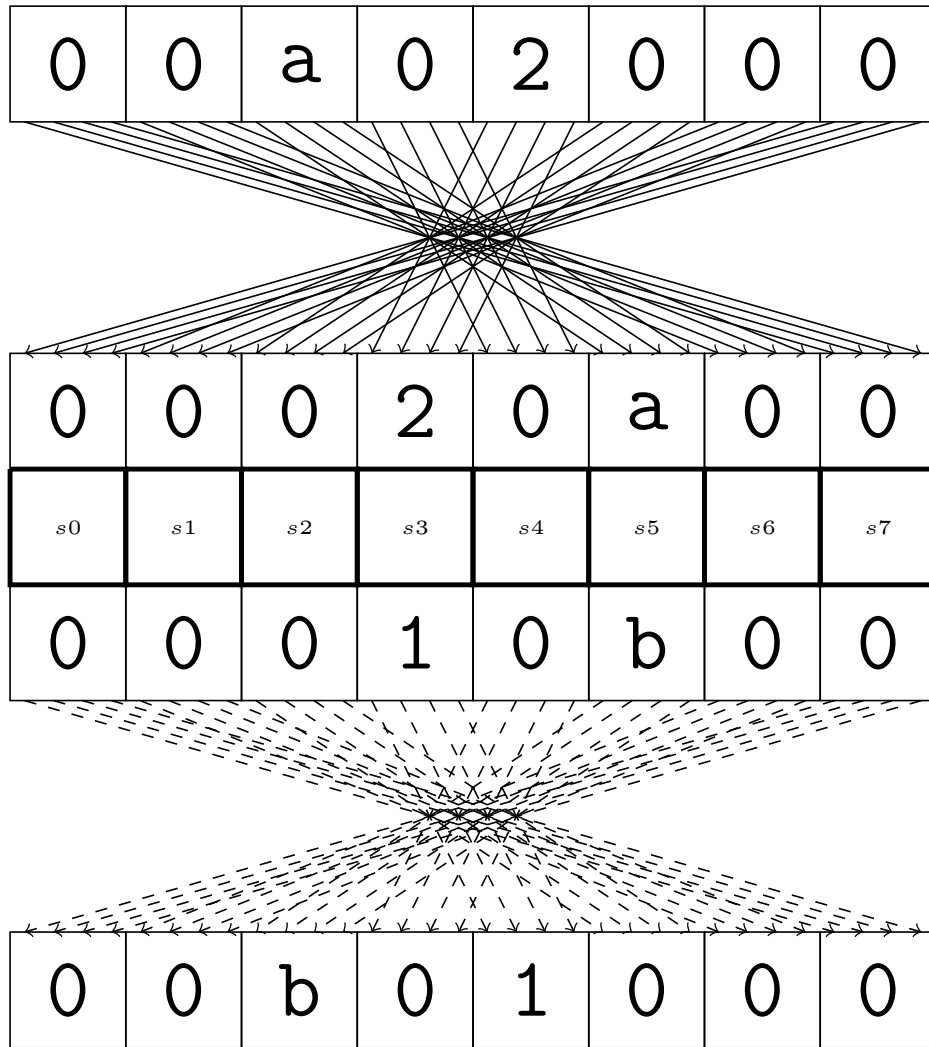


14 LBlock_round

Maximal differential probability: $2^{-4.0}$ 

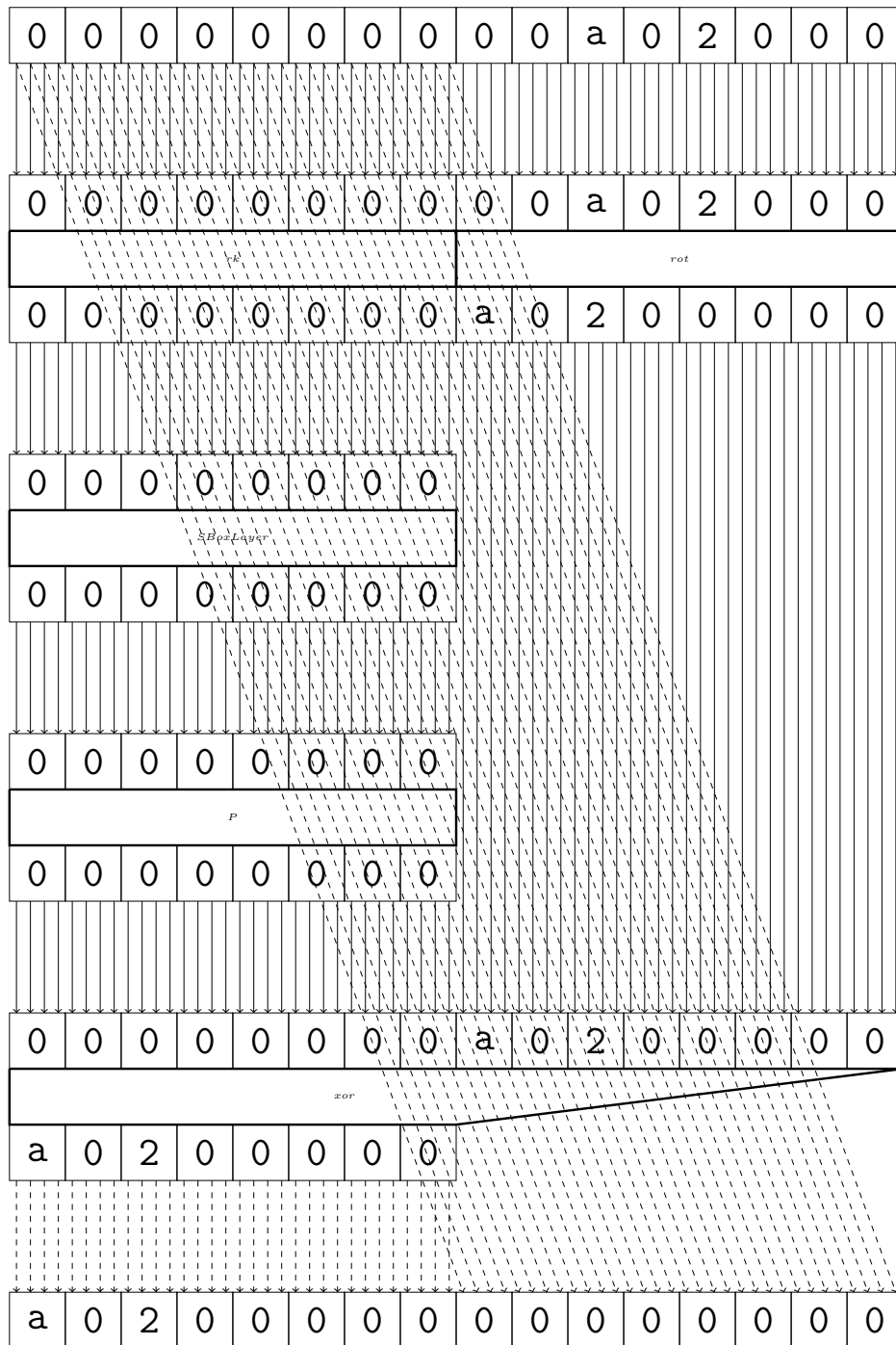
15 SBoxLayer

Maximal differential probability: $2^{-4.0}$



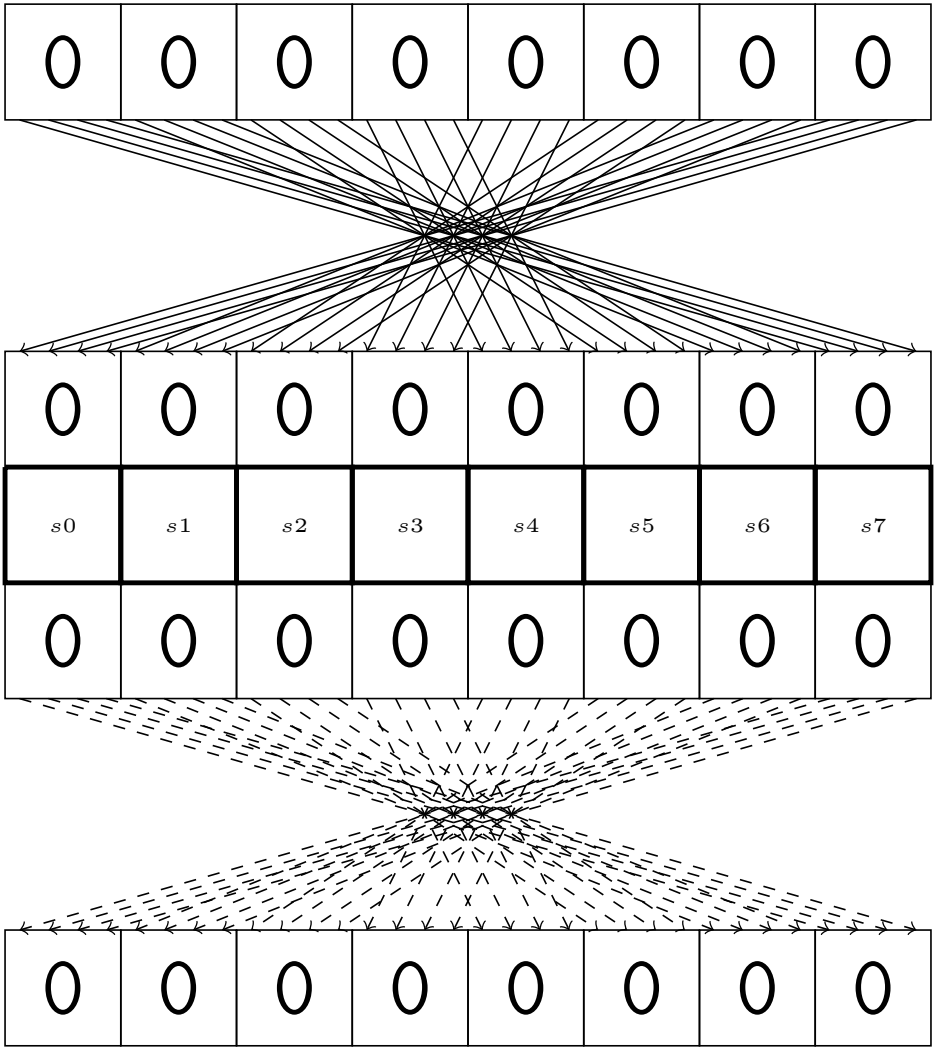
16 LBlock_round

Maximal differential probability: $2^{-0.0}$

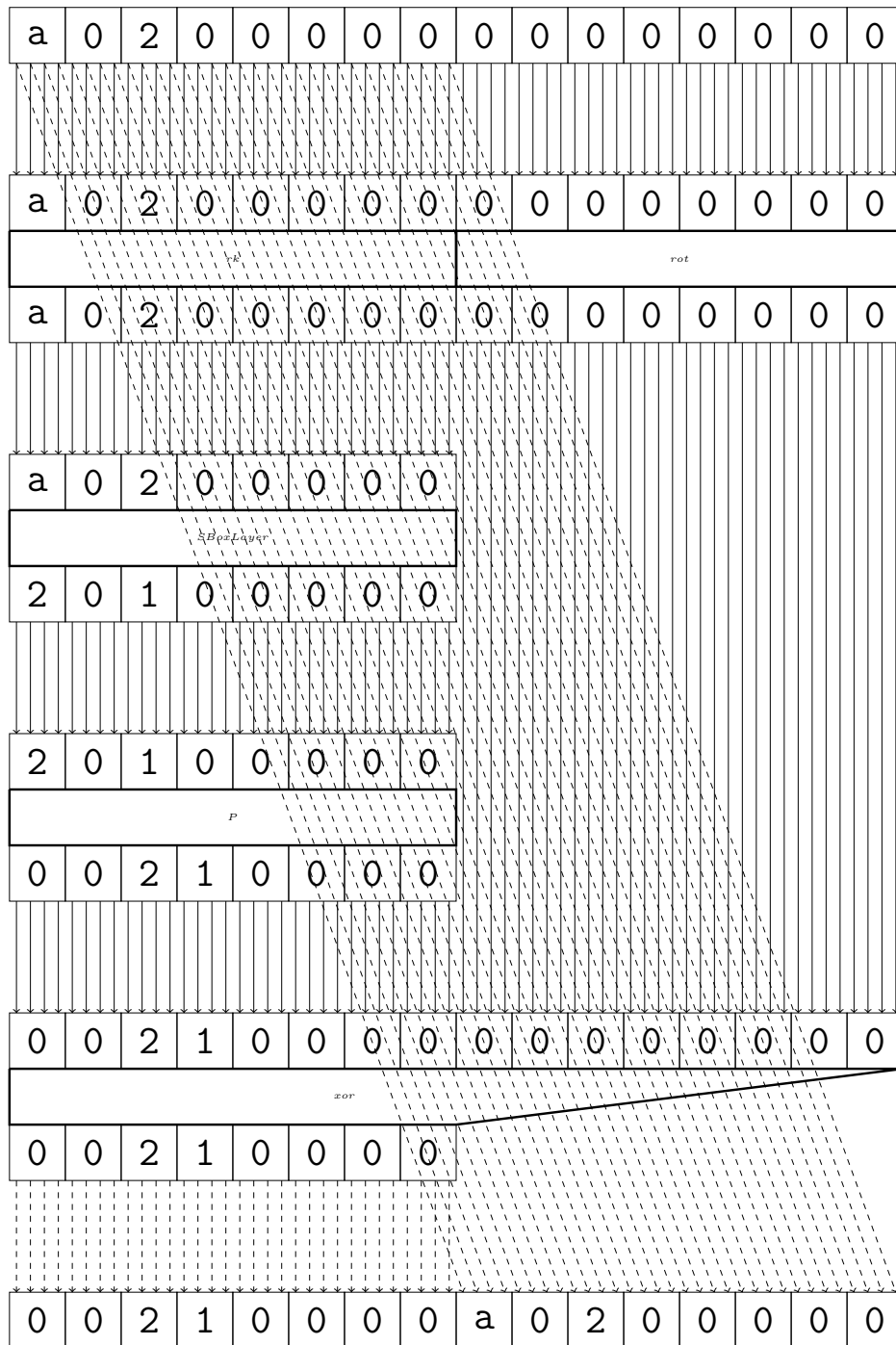


17 SBoxLayer

Maximal differential probability: $2^{-0.0}$

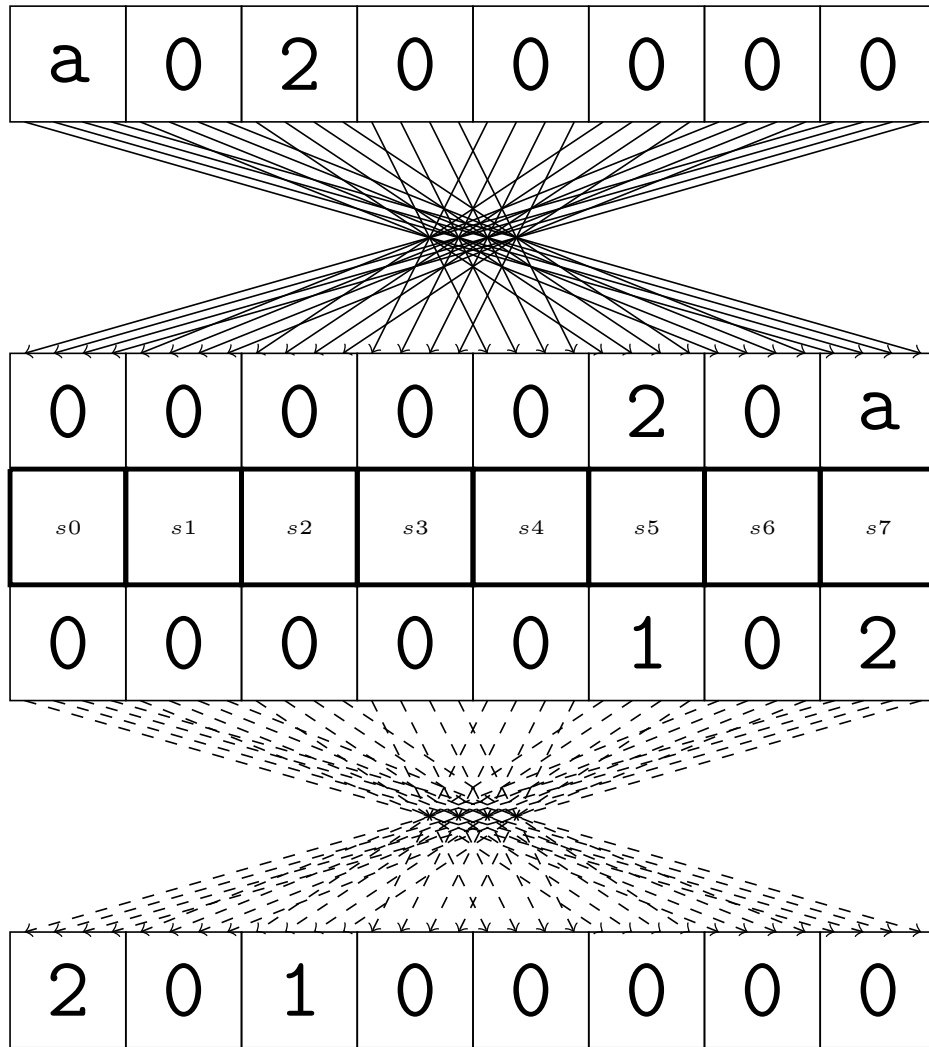


18 LBlock_round

Maximal differential probability: $2^{-4.0}$ 

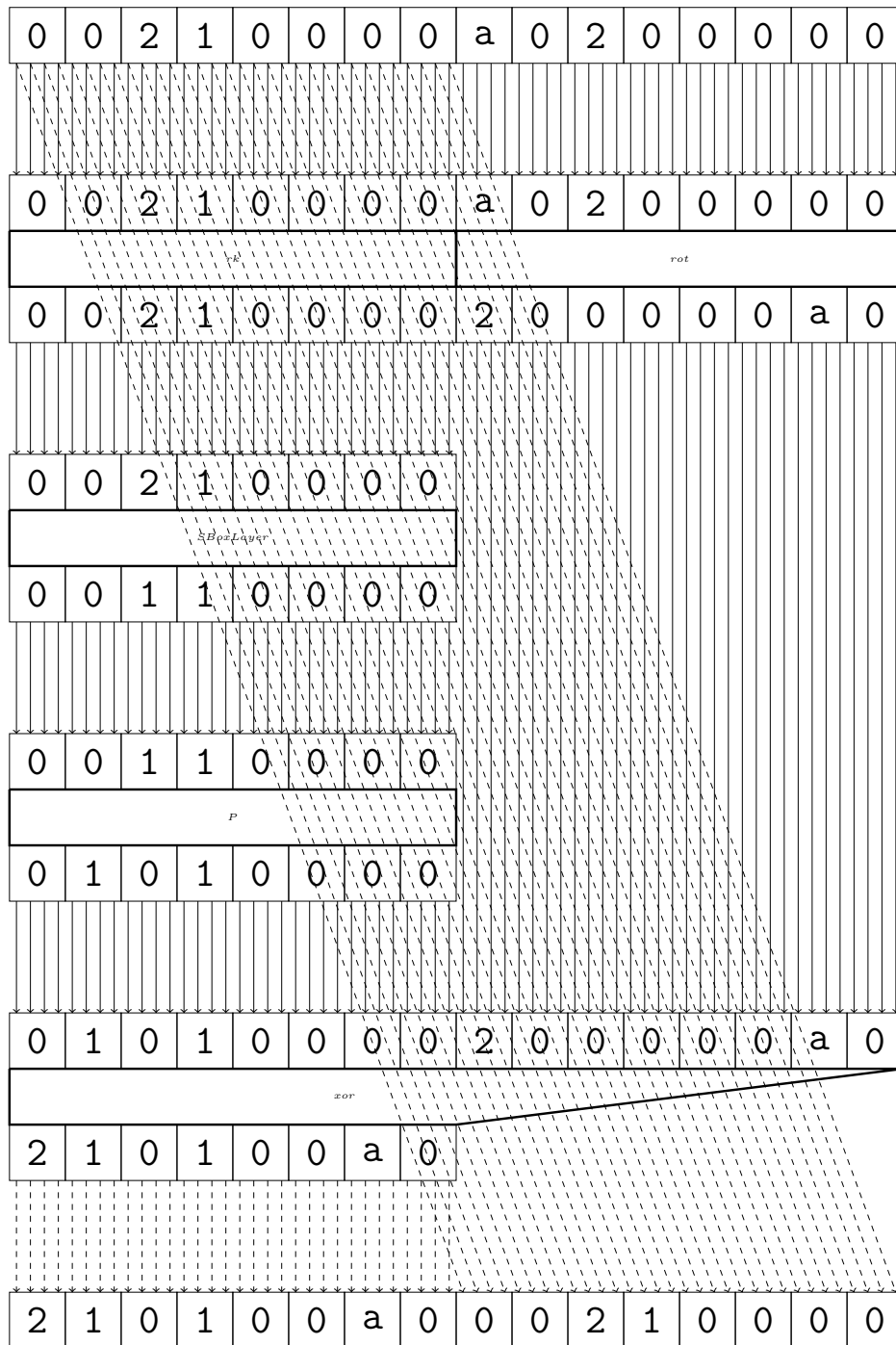
19 SBoxLayer

Maximal differential probability: $2^{-4.0}$



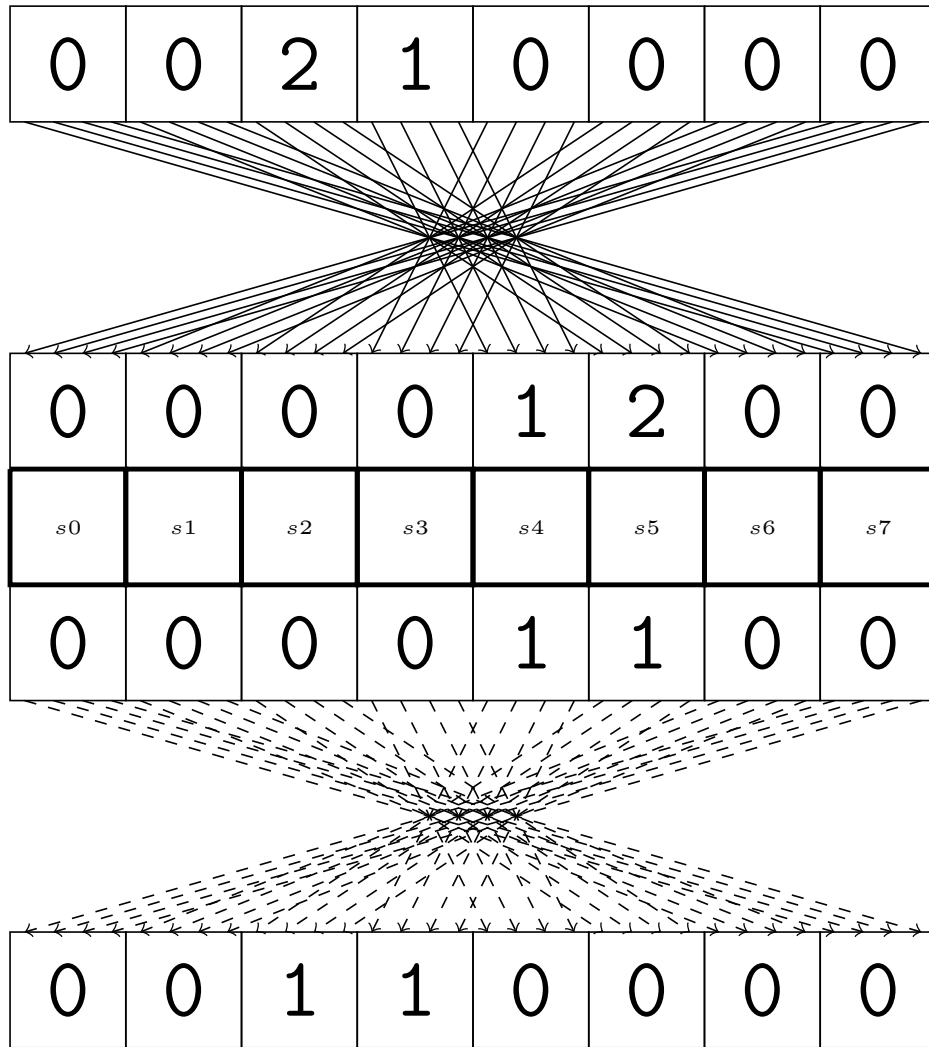
20 LBlock_round

Maximal differential probability: $2^{-4.0}$



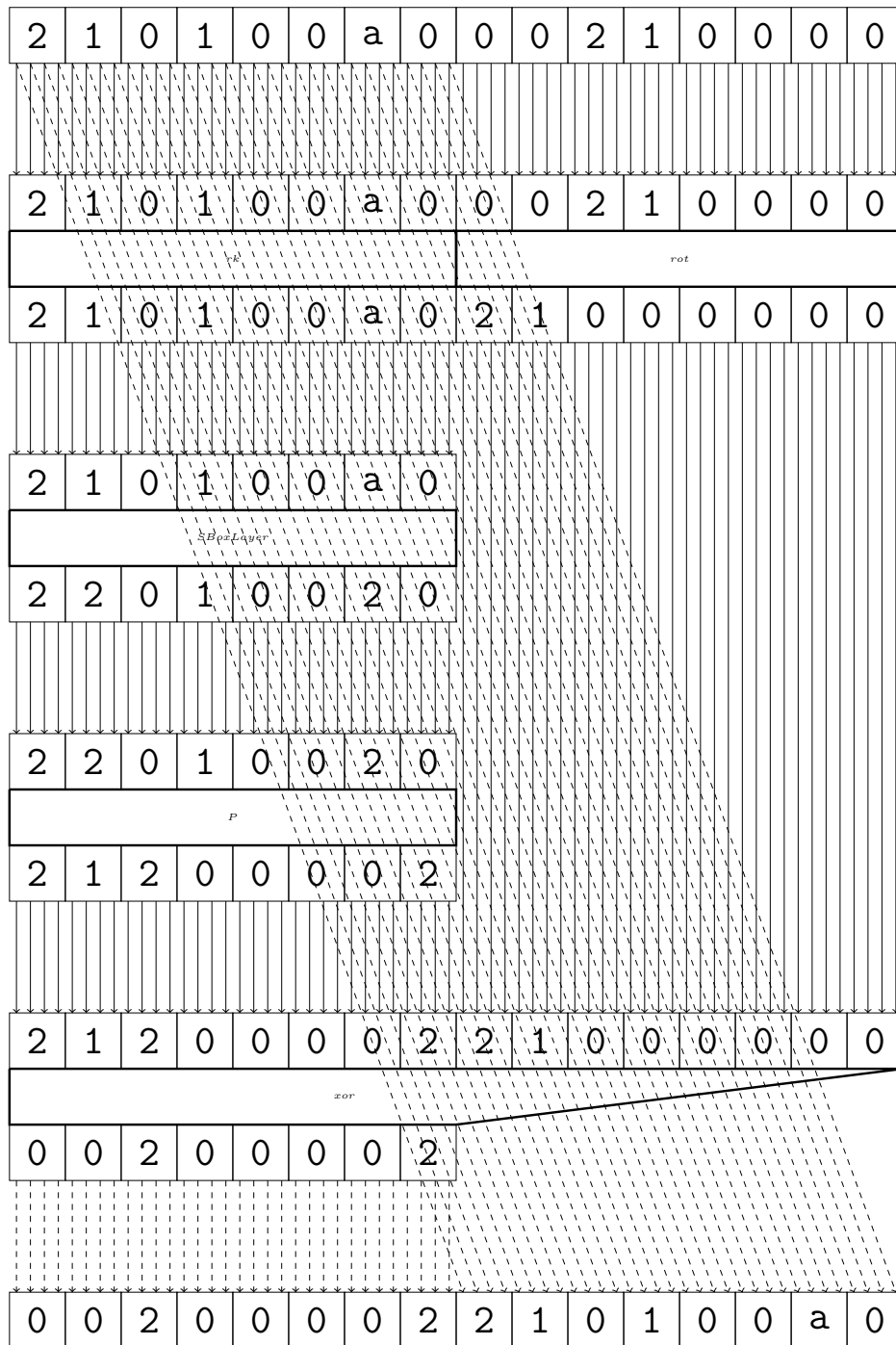
21 SBoxLayer

Maximal differential probability: $2^{-4.0}$



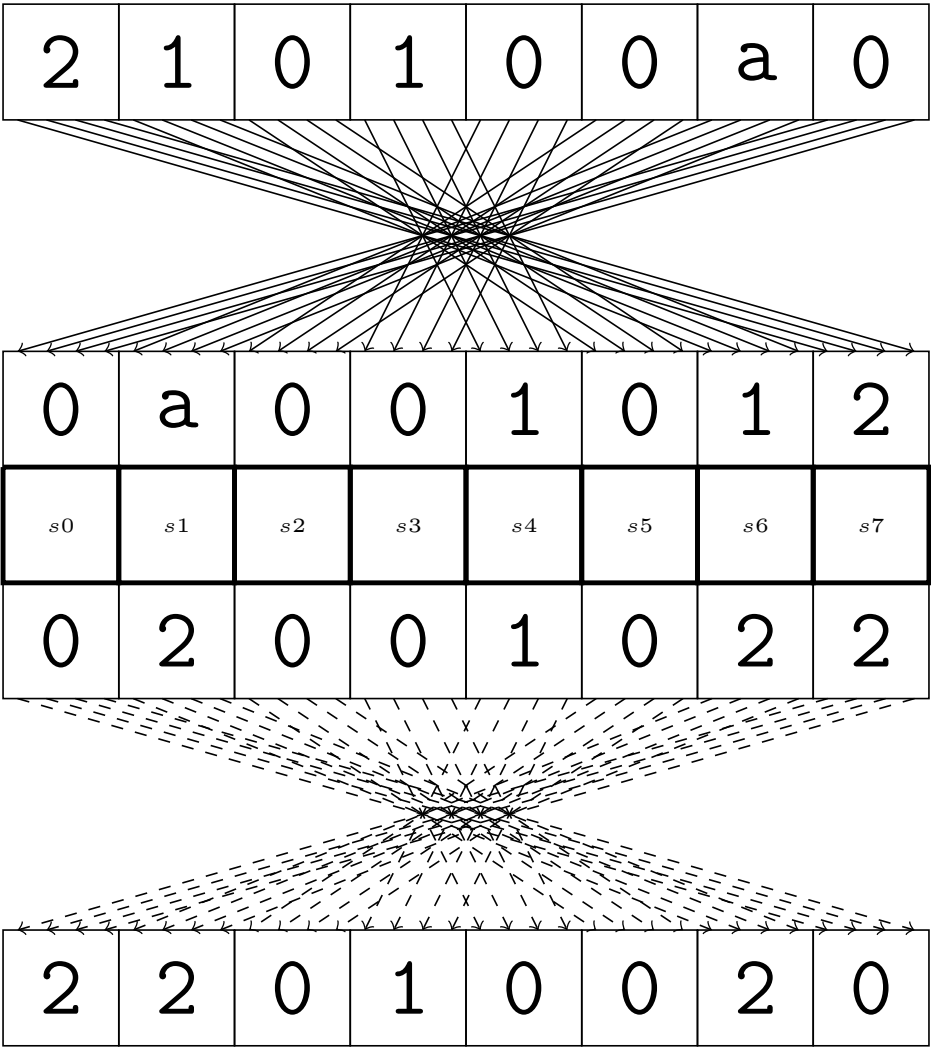
22 LBlock_round

Maximal differential probability: $2^{-8.0}$



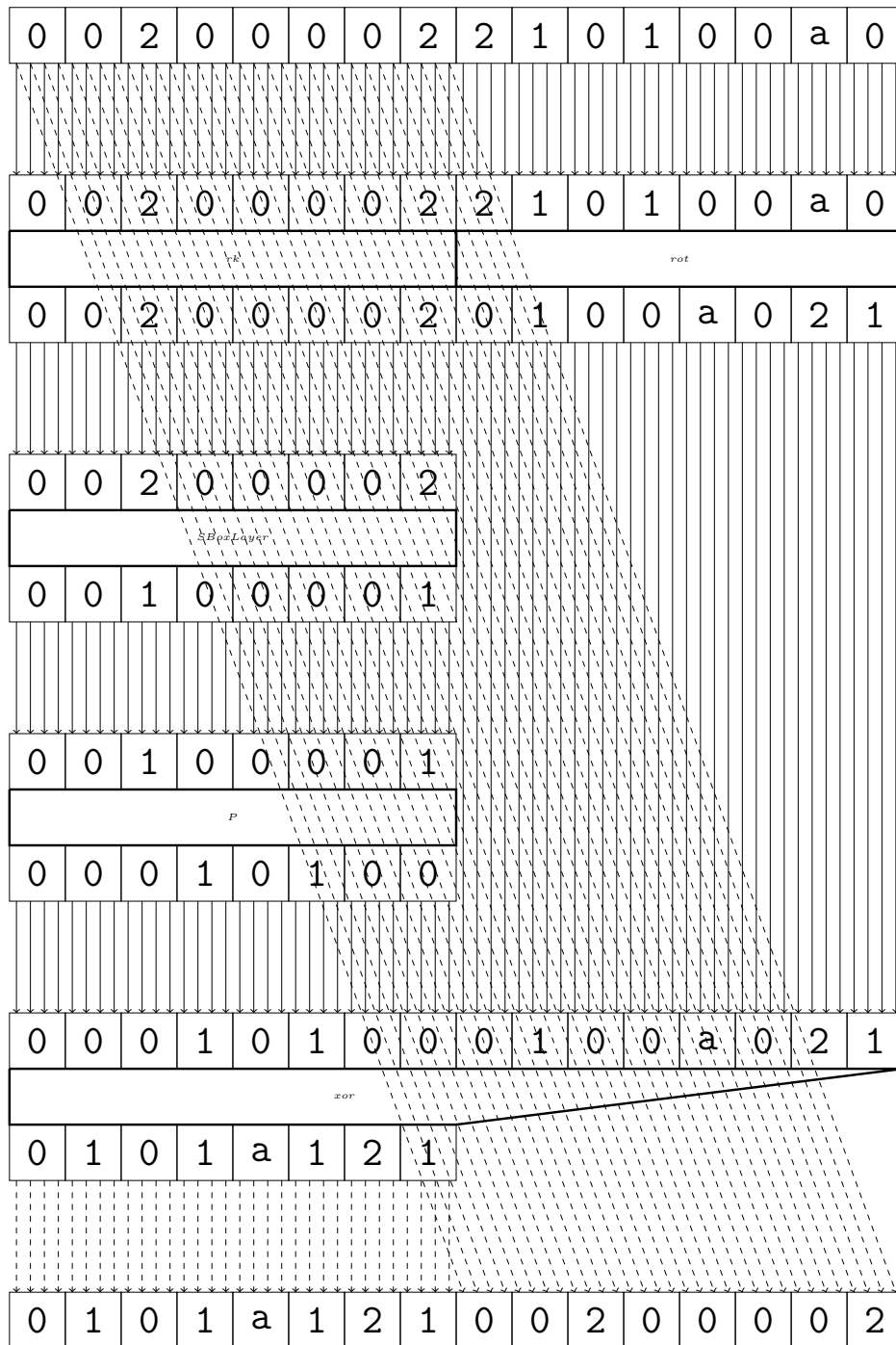
23 SBoxLayer

Maximal differential probability: $2^{-8.0}$



24 LBlock_round

Maximal differential probability: $2^{-4.0}$



25 SBoxLayer

Maximal differential probability: $2^{-4.0}$

